Predicting Diabetes from Health and Lifestyle Indicators

A Support Vector Machine Approach Using NHIS 2022 Data

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Introduction

Diabetes affects roughly **10% of US adults** and is strongly linked to modifiable lifestyle factors such as diet, physical activity, and weight. Identifying which factors most strongly predict diabetes status from population-level survey data can inform screening strategies and public health priorities.

This work uses **Support Vector Machine** classifiers on demographic and self-reported behavioral data from the **2022 National Health Interview Survey** (NHIS), accessed via IPUMS Health Surveys, to address the following:

Research Questions

1. Which predictors carry the most signal?
2. Does nonlinear flexibility improve over a linear baseline?
3. What do results suggest about prevention and screening?

Theoretical Background

SVMs find a decision boundary that separates two classes while **maximizing the margin** between the boundary and the closest training points (the *support vectors*). Only the support vectors influence the boundary.

Maximal Margin Classifier

$$\max M \ \text{ s.t. } \ y_i\left(\beta_0 + \sum_j \beta_j x_{ij}\right) \geq M, \ \ \sum_j \beta_j^2 = 1$$

Support Vector Classifier (soft margin)

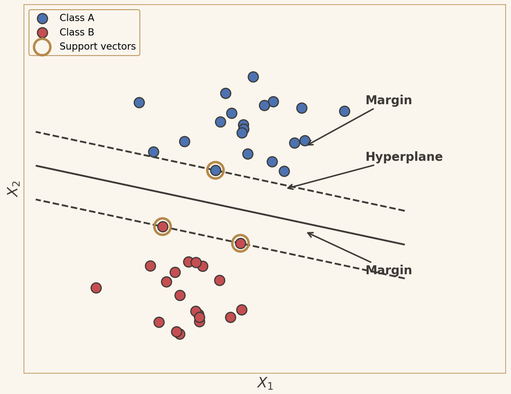
$$\max M \ \text{ s.t. } \ y_i\left(\beta_0 + \sum_j \beta_j x_{ij}\right) \geq M(1-\varepsilon_i), \ \ \varepsilon_i \geq 0, \ \ \sum_i \varepsilon_i \leq C$$

Kernelized Form

$$f(x) = \beta_0 + \sum_{i \in \mathcal{S}} \alpha_i K(x, x_i) \qquad \text{(only support vectors } \mathcal{S} \text{ contribute)}$$

Three Kernels Compared

Linear	RBF	Polynomial
$K(x, x') = x \cdot x'$	$K(x, x') = \exp(-\gamma\lVert x - x'\rVert^2)$	$K(x, x') = (1 + \gamma\, x \cdot x')^d$
Flat decision boundary	*Smooth nonlinear*	*Curved, fixed degree*



Conceptual SVM in 2D: solid line = hyperplane, dashed = margins.

Methodology

Data Preparation

NHIS 2022 subset to adults aged 18-85 (**n = 27,201** after cleaning). Special survey codes for unknown or out-of-universe responses were replaced with missing values per the codebook; complete cases retained. Target: ever diagnosed with diabetes (**10.3% prevalence**, 10:1 class imbalance).

Eight Predictors

BMI, age, sex, moderate and vigorous activity (min/day), sleep hours, soda consumption, income-to-poverty ratio.

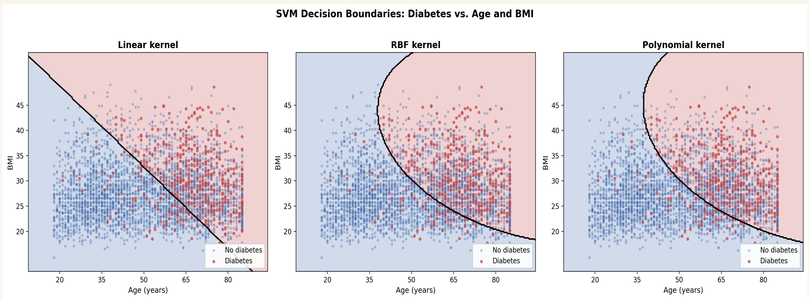
Model Fitting Strategy

01. 80/20 stratified train/test split (preserves prevalence)
02. Standardization (z-score, fit on train only) — SVMs are scale-sensitive
03. 3-fold cross-validation, scoring metric = ROC-AUC
04. class_weight='balanced' to compensate for the 10:1 imbalance

Hyperparameter Grids

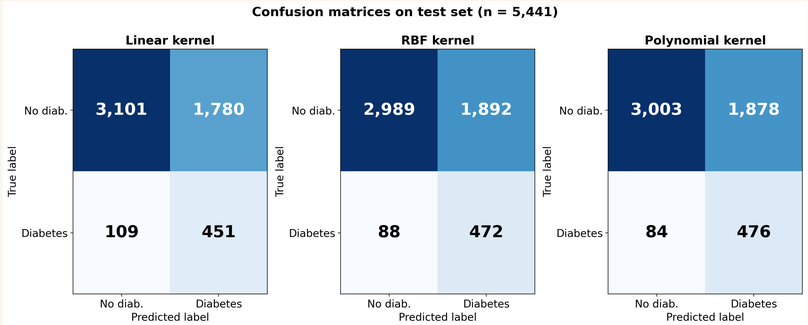
Kernel	C	Other parameters
Linear	0.1, 1, 10	—
RBF	0.5, 1, 2	γ ∈ {0.005, 0.01, 0.02}
Polynomial	0.1, 1, 10	deg ∈ {2, 3}, γ ∈ {scale, 0.01}

Decision Boundaries



Decision boundaries on AGE × BMI. Linear cuts the space with one straight line; RBF and polynomial bend around the high-AGE, high-BMI corner.

Confusion Matrices



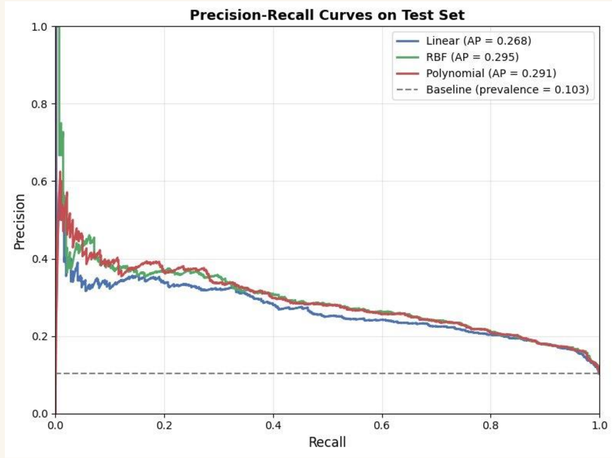
Test set (n = 5,441). All three kernels recover ~80-85% of true diabetics; ~37% of non-diabetics flagged — the high-recall profile expected for screening.

Results

Test-set evaluation on n = 5,441 held-out observations.

Kernel	AUC	AP	Precision	Recall	F1	# SVs (% train)
Linear	0.790	0.268	0.20	0.81	0.32	14,422 (66%)
RBF	0.804	0.295	0.20	0.84	0.32	14,050 (65%)
Polynomial	0.803	0.291	0.20	0.85	0.33	14,113 (65%)

AP (Average Precision) is more informative than AUC under heavy class imbalance; baseline AP at 10.3% prevalence = 0.103.



Precision-recall curves on the test set. AP values are 2.6–2.9× the no-skill baseline (0.103).

Discussion

Convergent kernels. All three reached AUC within 0.015 (0.790 / 0.804 / 0.803) and AP within 0.027 (0.268 / 0.295 / 0.291). The decision boundaries confirm the picture: linear cuts a single line, RBF and polynomial bend gently around the high-AGE, high-BMI corner, but the curvature does not improve AUC. The underlying risk surface is approximately linear in the observed range.

Recall over precision. With class_weight='balanced', each model captures 80–85% of true diabetics at the cost of flagging ~37% of non-diabetics (precision ~20%). AP values of 0.27–0.30 are roughly 3× the no-skill baseline of 0.103 — the models clearly learn signal — but absolute precision is bounded by 10:1 prevalence. The high-recall profile is the desired behavior for a screening tool.

Model complexity. Roughly 65% of training observations serve as support vectors across all three kernels — most points lie on or near the decision boundary. Combined with the kernel convergence, this points to substantial irreducible noise: overlapping risk distributions, self-report measurement error, and unobserved comorbidities limit how well any model can separate the classes from these inputs alone.

Limitations

- Cross-sectional design prevents causal claims
- AGE topcoded at 85 compresses the upper risk tail
- Self-reported lifestyle variables introduce error
- Sample skews higher-income (median ~3.75× poverty)

Conclusions

High recall (80–85%) with low precision (~20%) is the appropriate profile for screening. **Linear SVM is the recommended model** — equivalent AUC and AP to the kernelized models, simpler to interpret, no kernel evaluation at prediction time. **BMI and age** anchor the modifiable + non-modifiable risk axes; physical activity adds discriminative signal. **Age-stratified screening** (≥ 45) and population interventions targeting BMI and activity offer the highest expected impact.

References

[1] Blewett, L. A., Drew, J. A. R., King, M. L., et al. (2024). IPUMS Health Surveys: National Health Interview Survey, Version 7.4. Minneapolis, MN: IPUMS.

[2] James, G., Witten, D., Hastie, T., & Tibshirani, R. (2021). An Introduction to Statistical Learning, with Applications in Python (2nd ed.). Springer.

[3] Pedregosa, F. et al. (2011). Scikit-learn: Machine Learning in Python. JMLR, 12, 2825-2830.

[4] Centers for Disease Control and Prevention (2024). National Diabetes Statistics Report.

Code: github.com/brunamportoDS/diabetes-svm-classification